PROJECT PROPOSAL REPORT

FocusTrack: Multimodal Concentration Detection for Screen and Non-Screen Learners

Individual Component

Context-Aware Learner Interaction State Modeling for Concentration Assessment

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Figure 1:

ABSTRACT

Concentration is an important factor in effective learning, but it cannot always be inferred reliably from a learner's visible computer activity. University students perform different learning activities, including coding, reading, researching, writing assignments and examination preparation. Consequently, the same computer interaction behaviour may have different meanings depending on the learning activity. For example, low keyboard activity may represent focused reading or thinking, while frequent application switching may represent either legitimate information searching or distraction.

Previous research has demonstrated that keyboard, mouse and application interaction traces can provide information related to engagement, cognitive workload and task behaviour [1]–[6]. However, digital behaviour can be ambiguous when the learner's activity and temporal behaviour are not considered [2], [5]. This creates a need to investigate how learner interaction should be represented before it is used for concentration assessment.

This component proposes a Context-Aware Learner Interaction State Modeling approach. Timestamped keyboard, mouse and application metadata will be collected during predefined learning activities. The data will be transformed into temporal behavioural representations, from which recurring interaction states and transitions between states will be identified. Learning activity will then be incorporated as a contextual variable when interpreting these behavioural patterns.

The research will use a quantitative experimental design and Data Science techniques including temporal feature engineering, unsupervised state discovery, sequential modelling and supervised machine learning. A conventional aggregated-feature model will be used as a baseline. The proposed representation will be evaluated using participant-level validation, cross-context validation, ablation analysis and classification metrics, with Macro-F1 as the primary metric.

The component contributes the behavioural intelligence layer of FocusTrack. Its validated behavioural states, transitions and context-aware outputs will subsequently be aligned with environmental and visual information from the other components.

Keywords: Learner Behaviour, Context-Aware Modeling, Interaction States, Learning Analytics, Concentration Assessment

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TABLE OF CONTENTS

1. INTRODUCTION
   1.1 Background and Context
   1.2 Literature Review
   1.3 Existing Systems and Related Work
   1.4 Problem Statement
   1.5 Research Gap
   1.6 Research Questions
   1.7 Evidence-to-Decision Summary
2. OBJECTIVES
   2.1 Main Objective
   2.2 Specific Objectives
3. METHODOLOGY
   3.1 Research Methodology
   3.2 Data, Participants and Experimental Inputs
   3.3 Data Preparation
   3.4 Behavioural Representation
   3.5 Interaction-State Discovery
   3.6 State-Transition Modelling
   3.7 Context-Aware Behavioural Modelling
   3.8 Machine Learning
   3.9 Baseline / Benchmark
   3.10 Evaluation Metrics
   3.11 Validation Strategy
   3.12 Self-Evaluation Criteria
   3.13 Development Approach
   3.14 Ethical and Privacy Considerations
4. HIGH-LEVEL SYSTEM ARCHITECTURE
   4.1 Component Architecture
   4.2 Integration with FocusTrack
   4.3 Individual Responsibility Boundary

5. USER REQUIREMENTS
    5.1 Requirements-Elicitation Evidence
    5.2 Proposed User Requirements
    5.3 Functional Requirements
    5.4 Non-Functional Requirements
6. COMMERCIALIZATION PLAN
    6.1 Target Market
    6.2 Customer Personas
    6.3 Value Proposition
    6.4 Revenue Model
    6.5 Pricing Strategy
    6.6 Competitive Advantage
    6.7 Intellectual Property Considerations
7. BUDGET AND JUSTIFICATION
8. WORK BREAKDOWN STRUCTURE
9. GANTT CHART
10. REFERENCES
11. APPENDICES

LIST OF TABLES

Table 1: Literature Review Summary

Table 2: Existing Systems and Related Work

Table 3: Behavioural Data Sources

Table 4: Evaluation Metrics

Table 5: Individual Responsibility Boundary

Table 6: Proposed User Requirements

Table 7: Functional Requirements

Table 8: Non-Functional Requirements

Table 9: Budget and Justification

Table 10: Ten-Month Work Breakdown Structure

Table 11: Ten-Month Gantt Plan

Table 12: Behavioural Data Schema

Table 13: Evidence-to-Decision Log

LIST OF FIGURES

Figure 1: High-Level Component Architecture

Figure 2: FocusTrack Integration Architecture

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
DS	Data Science
EDA	Exploratory Data Analysis
HMM	Hidden Markov Model
RF	Random Forest
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting
WPM	Words Per Minute
F1	F1 Score
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SHAP	SHapley Additive exPlanations
IoT	Internet of Things
EdTech	Educational Technology

1. INTRODUCTION

1.1 Background and Context

While it is important to concentrate in learning, it is not always possible to infer the level of concentration from the visible computer activity of the learner. Students at the University engage in various learning activities such as writing assignments, reading text, researching and preparing for exams, coding activities, etc. Consequently, the same computer interaction behaviour may have different meanings depending on the learning activity. For instance, the number of times the keyboard is interacted with could be a sign of focused reading or thinking, and the amount of time it takes to switch applications could be a sign of information searching or distraction.

Previous studies show that traces of keyboard, mouse and application interaction can be used as information regarding engagement [1]–[6] and cognitive workload and task behaviour [5]. The interpretation of digital behavior, however, is not unambiguous if the learner's behavior and temporal behavior are not taken into account, however [2, 5]. This implies that it is necessary to explore how learner interaction should be represented prior to its use for assessment of concentration.

This component suggests an approach for Context-Aware Learner Interaction State Modeling. Meta data for the keyboard, mouse and/or applications will be collected and will be timestamped during certain learning activities. Data will be converted to temporal behavioural representations and then repeated states of interactions and transitions between states will be recognized. These behavioural patterns will then be interpreted in the context of learning activity.

The research will employ a quantitative experimental design with the application of Data Science techniques such as temporal feature engineering, unsupervised state discovery, sequential modelling and supervised machine learning techniques. An aggregated-feature model will be used as a baseline. We will validate the proposed representation by using participant level validation, cross-context validation, ablation analysis as well as classification metrics, with the primary being Macro-F1.

The component builds the behavioural intelligence part of FocusTrack. The validated behavioural states, transitions and context-awareness outputs will then be matched with the contextual and visual outputs provided by the other components.

The related keywords are: Learner Behaviour, Context-Aware Modeling, Interaction States, Learning Analytics, Concentration Assessment.

For example:

- low keyboard activity may indicate reading;
- low keyboard activity may indicate thinking;
- high keyboard activity may indicate productive coding;
- frequent application switching may indicate legitimate information searching;
- frequent application switching may also indicate distraction.

Therefore, behaviour should not be interpreted independently of the learner's activity.

Biedermann et al. [2] demonstrated the difficulty of distinguishing on-task and off-task behaviour from digital traces without contextual information. Learning-analytics research has also demonstrated the importance of contextual and instructional conditions when interpreting learner behaviour [5].

This creates a need for a behavioural modelling approach that considers not only what interaction occurred, but also:

- how the interaction changes over time;
- which behavioural patterns repeatedly occur;
- how one behavioural state changes into another;
- and what learning activity the learner is performing.

This component therefore investigates a context-aware temporal representation of learner interaction.

The proposed approach is:

```
Learning Context → Behavioural Events → Temporal Representation → Interaction States →
State Transitions → Context-Aware Representation → Concentration-Related Prediction
```

The component focuses on three low-cost behavioural sources:

- keyboard interaction;
- mouse interaction;
- application interaction.

Raw key characters, passwords, messages and clipboard contents will not be collected.

1.1.1 Real-World Problem

Students currently have limited objective information about how their behaviour changes during learning sessions.

Many existing productivity systems focus primarily on:

- time spent;
- application usage;
- website usage;
- blocking distracting applications;
- or simple productivity scores.

Such measurements do not necessarily explain why the learner's interaction changed.

A student may spend ten minutes without typing because they are reading a textbook, analysing a problem or thinking about a solution. Therefore, interpreting this period simply as “inactive” can be misleading.

The real-world problem is consequently:

Students need concentration-related information that considers the context and temporal structure of their learning behaviour rather than relying only on simple activity totals.

1.1.2 Affected Users

The primary users are university students participating in learning activities.

Secondary stakeholders include:

- university learning-support services;
- academic researchers;
- higher-education institutions;
- EdTech developers.

The overall FocusTrack system is intended to support both screen-intensive and non-screen learning behaviour. The behavioural component specifically provides information when digital interaction is available and contributes its output to the overall multimodal system.

1.1.3 Proposed Solution

The proposed component will not interpret each interaction event independently.

Instead, it will construct a temporal behavioural representation. For example:

```
Individual Events
    ↓
Temporal Features
    ↓
Behavioural Patterns
    ↓
Interaction States
    ↓
State Transitions
    ↓
Learning Context
    ↓
Context-Aware Behavioural Representation
    ↓
Concentration-Related Evidence
```

The objective is to investigate whether this richer representation provides additional useful information compared with conventional aggregated features.

1.1.4 Sustainable Development Goal Alignment

The overall FocusTrack project contributes primarily to United Nations Sustainable Development Goal 4: Quality Education by investigating data-driven approaches that can support learning effectiveness and personalized educational support.

This component contributes to that goal by investigating a low-cost behavioural analytics approach that can provide additional information about learner interaction during study sessions.

1.1.5 Research Cluster Alignment

The research aligns with the Data Science / Artificial Intelligence research area within the FocusTrack project's Smart Systems, IoT and Artificial Intelligence direction.

The main technical pillars used by this component are:

- Data Science;
- Machine Learning;
- behavioural analytics;
- temporal data analysis;
- unsupervised learning;
- supervised learning.

1.2 LITERATURE REVIEW

The literature review focuses on five questions:

1. Can computer interaction provide useful behavioural information?

2. Are individual behavioural features reliable indicators of internal states?

3. Does context affect the interpretation of behaviour?

4. Does temporal interaction contain information beyond aggregate features?

5. Can recurring behavioural states be discovered from interaction traces?

1.2.1 Behavioural Engagement Detection

Altuwairqi et al. [1] investigated student behaviour for measuring engagement in online learning environments using interaction and additional behavioural signals.

Method: The study combined computer interaction information with other behavioural information to classify learner engagement.

Contribution: The research provides evidence that low-cost computer interaction can contain useful information related to learner engagement.

Strength: It demonstrates the feasibility of using interaction behaviour as a source of information for learner-state analysis.

Limitation: Engagement modelling does not necessarily mean that individual interaction features have a fixed interpretation across different learning activities.

Influence on Proposed Research: This research supports the use of keyboard and mouse metadata as behavioural inputs but motivates the investigation of a richer context-dependent representation.

1.2.2 Contextualized Logging of Learning Behaviour

Biedermann et al. [2] investigated contextualized logging of on-task and off-task behaviour during learning.

Method: Digital learning traces were analysed together with contextual information concerning learner behaviour.

Finding: Digital traces alone can be insufficient for determining whether an observed activity is genuinely on-task or off-task.

Strength: This study directly supports the importance of contextual interpretation.

Limitation: The contextual information does not directly establish the proposed interaction-state representation for concentration assessment.

Influence: This study provides important evidence for treating learning activity as an explicit contextual variable rather than interpreting interaction behaviour in isolation.

1.2.3 Mouse Behaviour and Cognitive Workload

Cha and Min [3] investigated the relationship between unconscious mouse behaviour and cognitive workload.

Freihaut and Göritz [4] reviewed the use of mouse behaviour for stress measurement.

Contribution: These studies demonstrate that mouse movement characteristics can contain information associated with cognitive or affective conditions.

Strength: Mouse interaction is non-invasive and can be collected without specialised physiological equipment.

Limitation: Mouse behaviour is not a universal or standalone indicator of concentration. The relationship between mouse behaviour and internal states can vary depending on experimental conditions.

Influence: Mouse features will therefore be treated as one behavioural signal among several rather than being directly mapped to concentration.

1.2.4 Context in Learning Analytics

Jovanović et al. [5] investigated how learner-related and instructional conditions influence learning analytics predictions.

Contribution: The study supports the importance of considering contextual and learner conditions when modelling educational behaviour.

Limitation: The research focuses on academic outcomes rather than session-level concentration.

Influence: The proposed component extends the contextual principle to the interpretation of temporal computer interaction during individual learning sessions.

1.2.5 Task Switching and Interaction Sequences

Meyer et al. [6] investigated the detection of developer task switches using interaction traces.

Contribution: The research demonstrates that interaction sequences and application changes can provide information about task-state transitions.

Strength: It shows why a sequence of events can provide more information than a simple count.

Limitation: The research focuses on software-development tasks and therefore does not directly address general university learning activities.

Influence: The proposed component will investigate whether similar temporal state and transition representations can be useful in broader learning contexts.

1.2.6 Behavioural Phase Discovery

Cavallin and Spagnol [8] investigated behavioural traces during GenAI-supported creative work and identified recurring behavioural phases.

Contribution: The study provides recent evidence that recurring behavioural phases can be identified from interaction traces rather than completely predefined in advance.

Limitation: The study concerns human–AI creative work rather than student learning.

Influence: This supports investigating data-driven interaction-state discovery in the proposed component.

1.2.7 Multimodal Cognitive Assessment

Debie et al. [9] reviewed multimodal approaches for objective cognitive-workload assessment.

Contribution: The review supports the use of multiple information sources when analysing internal cognitive conditions.

Limitation: Many multimodal approaches involve physiological signals that can be intrusive or difficult to deploy in ordinary learning environments.

Influence: Component 1 provides a low-cost behavioural modality that can later be combined with environmental and visual information in FocusTrack.

1.2.8 Literature Synthesis

Research	Main Signal / Method	Main Finding	Limitation	Influence
Altuwairqi et al. [1]	Keyboard, mouse and additional behavioural signals	Interaction contains engagement-related information	Activity context not central	Use behavioural interaction as input
Biedermann et al. [2]	Digital learning traces + context	Digital behaviour can be ambiguous	Contextualisation remains challenging	Include learning context
Cha & Min [3]	Mouse behaviour	Mouse behaviour relates to workload	Not a standalone concentration measure	Use mouse as one signal
Freihaut & Göritz [4]	Mouse/stress studies	Mouse behaviour has measurable relationships	Reliability varies	Avoid single-feature interpretation
Jovanović et al. [5]	Learning analytics + contextual conditions	Context improves interpretation/prediction	Focuses on academic outcomes	Condition behavioural modelling on context
Meyer et al. [6]	Desktop/IDE sequences	Sequences reveal task transitions	Developer-specific	Model behavioural transitions
Cavallin & Spagnol [8]	Behavioural traces	Recurring phases can be discovered	Different domain	Investigate state discovery
Debie et al. [9]	Multimodal workload	Multiple modalities can complement each other	Often uses physiological signals	Supply behavioural modality to FocusTrack

1.3 EXISTING SYSTEMS AND RELATED WORK

Existing systems can be broadly divided into:

1. productivity/time-tracking tools;

2. distraction-blocking tools;

3. research-based engagement systems;

4. interaction-sequence analysis systems.

Table 2. Existing Systems

System	Function	Main Approach	Limitation Relevant to Component
RescueTime	Application and website tracking	Application usage logging and categorisation	Does not explicitly model learner learning context

System	Function	Main Approach	Limitation Relevant to Component
Focus/blocking applications	Prevent access to distractions	Timers and allow/block rules	Prevention rather than behavioural modelling
Webcam-based engagement systems	Estimate engagement	Computer vision + interaction	Camera-dependent and not focused on context-conditioned interaction states
IDE task-switch analysis	Detect task switches	Interaction sequences	Mainly software-development specific

The current proposal should not claim that no existing system uses keyboard, mouse or application data.

The stronger argument is that the reviewed systems do not establish the specific representation investigated here:

Temporal interaction states + transitions + learning-context conditioning for concentration assessment.

This distinction makes the research gap more defensible.

1.4 PROBLEM STATEMENT

Computer interaction logs provide large quantities of behavioural information. However, raw interaction events do not directly represent concentration.

The same observable behaviour may correspond to different learner states. For example:

- low interaction may represent reading;
- low interaction may represent thinking;
- high interaction may represent productive coding;
- frequent application switching may represent research;
- frequent application switching may represent distraction.

Conventional aggregated features such as:

- typing rate;
- mouse activity;
- idle duration;
- application-switch count

may therefore lose information about the temporal structure of learner behaviour. They do not necessarily represent:

- how long a behavioural pattern persists;
- what state occurred before it;
- what state followed it;
- whether an interruption occurred;
- whether the learner recovered;
- whether the same pattern occurred during a different learning activity.

Therefore, the individual research problem is:

How can learner computer interaction behaviour be represented as temporal behavioural states and transitions, and how can the interpretation of these states be conditioned on learning context to provide more meaningful evidence for concentration assessment?

1.5 RESEARCH GAP

Previous research demonstrates that computer interaction traces can provide useful information about engagement, workload and task behaviour [1], [3], [4], [6].

Research on contextualized learning behaviour demonstrates that digital traces can be ambiguous without contextual information [2], while learning analytics research supports incorporating contextual conditions [5].

Research on interaction sequences further demonstrates that behavioural transitions can provide information about changing task states [6], and recent behavioural-trace research demonstrates that recurring behavioural phases can be discovered from interaction data [8].

However, based on the literature reviewed for this component, there is limited investigation of a unified representation that:

1. converts learner computer interaction into recurring temporal behavioural states;

2. models transitions between those states;

3. conditions their interpretation on the learner's learning activity; and

4. evaluates whether this representation provides additional concentration-related evidence beyond conventional aggregated behavioural features.

Therefore, the research gap is:

Existing research provides evidence that computer interaction traces are informative and that contextual and temporal information are important, but limited research has investigated a context-conditioned interaction-state representation that models behavioural states and transitions specifically for session-level concentration assessment across different learning activities.

The proposed research will empirically test this gap rather than claim that no related system exists.

1.6 RESEARCH QUESTIONS

RQ1

Does a temporal interaction-state representation provide more useful concentration-related evidence than conventional aggregated keyboard, mouse and application features?

RQ2

Does conditioning interaction states and state transitions on the learner's learning activity improve the generalisability of concentration prediction across different learning contexts?

1.7 EVIDENCE-TO-DECISION SUMMARY

Claim	Evidence	Decision
Computer interaction contains behavioural information	[1], [3], [4]	Use keyboard, mouse and application metadata
Individual behaviour can be ambiguous	[2], [4]	Avoid direct single-feature concentration rules
Context influences interpretation	[2], [5]	Include learning activity as an explicit variable
Temporal sequences contain state-change information	[6]	Model behavioural sequences and transitions
Recurring behavioural phases can be discovered	[8]	Investigate data-driven state discovery
Multiple modalities can complement each other	[9]	Provide behavioural outputs for FocusTrack multimodal fusion
Proposed contribution must outperform a simpler representation	Comparative experimental requirement	Establish aggregated-feature baseline and conduct ablation/cross-context evaluation

2. OBJECTIVES

2.1 Main Objective

To develop and empirically evaluate a context-aware learner interaction-state representation that transforms keyboard, mouse and application behaviour into temporal states and transitions conditioned on learning activity for concentration assessment.

2.2 Specific Objectives

SO1

To design and implement a privacy-preserving behavioural logging pipeline that records timestamped keyboard, mouse and application metadata.

SO2

To preprocess behavioural events and engineer temporal features and sequences representing interaction intensity, timing, inactivity and application transitions.

SO3

To discover recurring learner interaction states from temporal behavioural representations using suitable data-driven techniques.

SO4

To model and quantify transitions, persistence, interruptions and recovery patterns between interaction states.

SO5

To investigate how the meaning and predictive value of interaction states and transitions vary across different learning contexts.

SO6

To compare the proposed context-aware state-and-transition representation with a conventional aggregated-feature baseline using participant-level and cross-context validation.

3. METHODOLOGY

3.1 Research Methodology

The proposed research will use a quantitative experimental research design supported by Data Science and Machine Learning techniques.

This design is appropriate because the research questions require:

- measurable behavioural data;
- predefined learning activities;
- temporal observations;
- computational modelling;

- experimental comparison;
- and quantitative evaluation.

The research process will be:

```
Literature Review
    ↓
Research Gap
    ↓
Research Design
    ↓
Behavioural Data Collection
    ↓
Data Preparation
    ↓
Temporal Representation
    ↓
Interaction-State Discovery
    ↓
State-Transition Modelling
    ↓
Context-Aware Representation
    ↓
Baseline Model
    ↓
Proposed Model
    ↓
Evaluation
    ↓
Validation
    ↓
Statistical Analysis
    ↓
Integration
```

Importantly, data collection is the measurement stage and is not itself the primary research contribution.

3.2 DATA, PARTICIPANTS AND EXPERIMENTAL INPUTS

3.2.1 Participants

The overall FocusTrack TAF proposes approximately 20–25 university student participants, including screen-intensive and non-screen learners.

The final recruitment procedure, participant characteristics and session structure will follow the approved research protocol and ethics requirements.

3.2.2 Learning Activities

Potential learning activities include:

- coding;
- reading;
- assignment work;
- examination preparation;
- academic research.

The final activities will be determined according to the approved FocusTrack protocol.

The learning activity will be selected or recorded before each session.

This is important because behavioural meaning depends on the activity.

3.2.3 Behavioural Data

Keyboard

The system may capture:

- key event timestamps;
- inter-keystroke intervals;
- typing activity;
- typing bursts;
- backspace/error-related activity.

Raw key characters will not be stored.

Mouse

The system may capture:

- movement events;
- movement distance;
- velocity;
- acceleration;
- click activity;
- inactivity periods.

Application

The system may capture:

- active application category;
- application duration;
- application transitions;
- number of application switches;
- application usage sequence.

Temporal

The system will maintain:

- timestamps;
- session duration;
- idle duration;
- temporal-window duration;
- behavioural-state duration;
- state-transition information.

3.2.4 Concentration Reference

The overall FocusTrack TAF proposes collecting self-reported focus ratings after each session as the operational reference for concentration modelling.

These ratings should be treated as a reference measure rather than perfect objective ground truth.

The limitation of self-report will therefore be acknowledged when interpreting the final results.

3.3 DATA PREPARATION

Step 1 — Data Cleaning

Remove or flag:

- duplicate events;
- invalid timestamps;
- corrupted records;
- impossible values;
- incomplete records.

Step 2 — Timestamp Synchronisation

All behavioural events will be ordered using timestamps.

Step 3 — Session Segmentation

Data will be separated according to:

- participant;
- session;
- learning activity.

Step 4 — Temporal Windowing

The behavioural stream will be divided into suitable temporal windows. Fixed windows may initially be investigated, while adaptive or alternative windowing strategies may be considered if supported by pilot results.

Step 5 — Feature Extraction

Temporal behavioural features will be calculated.

Step 6 — Sequence Construction

Temporal windows will be converted into ordered behavioural sequences for state and transition analysis.

3.4 BEHAVIOURAL REPRESENTATION

The component will represent behaviour at three levels.

Level 1 — Individual Events

Examples:

- key event;

- mouse movement;
- mouse click;
- application change.

Level 2 — Temporal Behavioural Patterns

Events will be transformed into patterns such as:

- sustained interaction;
- low-interaction period;
- mouse-intensive activity;
- rapid application transitions;
- prolonged inactivity.

Level 3 — Interaction States

Recurring behavioural patterns will be represented as interaction states.

Potential states include:

- sustained interaction;
- low-interaction period;
- exploratory interaction;
- application transition;
- fragmented interaction;
- interruption;
- recovery.

These are candidate behavioural descriptions, not predefined concentration labels.

A state will not automatically be called “focused” or “distracted.”

Its relationship with concentration will be investigated empirically.

3.5 INTERACTION-STATE DISCOVERY

The research will investigate suitable unsupervised or semi-structured approaches for discovering recurring behavioural states.

Candidate methods include:

Clustering

Useful for identifying groups of similar behavioural feature patterns.

Gaussian Mixture Models

Useful when behavioural states may overlap probabilistically rather than forming completely separate groups.

Hidden Markov Models

Useful when behavioural observations are expected to evolve through hidden states over time.

Sequential Pattern Analysis

Useful for identifying recurring sequences of behavioural events or states.

The final method will be selected based on:

- suitability for the collected data;
- temporal capability;
- interpretability;
- computational feasibility;
- state stability;
- validation results.

3.6 STATE-TRANSITION MODELLING

Learner behaviour changes throughout a learning session.

Therefore, the research will model how learners move between states. For example:

```
State A
  ↓
State B
  ↓
State C
  ↓
State A
```

The following measures will be investigated:

- state duration;
- state frequency;
- transition frequency;
- transition probability;
- repeated transitions;
- number of state changes;
- interruption patterns;
- recovery patterns.

For example:

```
Sustained Interaction
      ↓
Application Transition
      ↓
Exploratory Interaction
      ↓
Sustained Interaction
```

may represent a different behavioural trajectory from:

```
Sustained Interaction
      ↓
Application Transition
      ↓
```

```
Application Transition
      ↓
Fragmented Interaction
```

even if both sessions contain a similar number of application switches.

This demonstrates why transition information may provide additional information beyond simple feature totals.

3.7 CONTEXT-AWARE BEHAVIOURAL MODELLING

Learning context will be incorporated into the behavioural representation.

The proposed analytical input is:

Learning Context + Temporal Behaviour + Interaction States + State Transitions

rather than:

Behaviour → Concentration

For example:

```
Coding
+
Sustained Interaction
+
Stable Transition Pattern
↓
Contextual Behavioural Evidence
```

whereas:

```
Reading
+
Low Keyboard Activity
+
Stable State
↓
Contextual Behavioural Evidence
```

The system therefore does not automatically interpret low keyboard activity as distraction.

The research will test whether incorporating context improves prediction and generalisation.

3.8 MACHINE LEARNING

Machine Learning will be applied after the behavioural representation has been constructed.

Candidate models include:

- Random Forest;
- XGBoost;
- Support Vector Machine.

These models are appropriate candidates for engineered behavioural features and can be evaluated systematically.

The final model will be selected according to:

- predictive performance;
- generalisation;
- computational feasibility;
- stability;
- interpretability.

The research will avoid selecting the model solely according to training accuracy.

3.9 BASELINE / BENCHMARK

A conventional behavioural model will be established as the baseline.

Baseline Features

The baseline will use:

- typing rate;
- mouse activity;
- idle duration;
- application-switch frequency;
- interaction frequency.

These features will be aggregated without explicit interaction-state or transition representation.

Proposed Model

The proposed representation will include:

- temporal behavioural features;
- interaction states;
- state transitions;
- learning context.

Therefore:

```
BASELINE
Aggregated Features
      ↓
Machine Learning
      ↓
Concentration Prediction
```

will be compared with:

```
PROPOSED
Temporal Features
      ↓
Interaction States
      ↓
State Transitions
      ↓
Learning Context
      ↓
Machine Learning
      ↓
Concentration Prediction
```

This directly tests whether the proposed representation provides value beyond conventional behavioural aggregation.

3.10 EVALUATION METRICS

Metric	Purpose
Accuracy	Overall prediction correctness
Precision	Correctness of predicted classes
Recall	Ability to identify relevant classes
Macro-F1	Primary balanced performance measure
Balanced Accuracy	Performance under class imbalance
ROC-AUC	Class discrimination where applicable

Primary Metric

Macro-F1 will be the primary metric because concentration-related classes may not be equally represented.

The study will not define an arbitrary success criterion such as “90% accuracy” before data collection.

Instead, success will be determined through comparison with the baseline.

3.11 VALIDATION STRATEGY

3.11.1 Participant-Level Validation

Data from the same participant should not appear in both training and test sets within an evaluation split.

This reduces participant-specific leakage.

3.11.2 Cross-Validation

Where the final dataset size permits, appropriate cross-validation will be applied.

3.11.3 Cross-Context Validation

The model will be evaluated across different learning activities.

This tests whether the behavioural representation generalises beyond one specific activity.

3.11.4 Ablation Analysis

The contribution of individual elements will be investigated by removing:

1. learning context;

2. interaction states;

3. state transitions.

This allows the research to determine where performance improvements originate.

3.11.5 Statistical Analysis

Performance will be analysed across participant-level evaluation folds.

Where appropriate, the final analysis will report:

- confidence intervals;
- effect sizes;
- statistical significance;
- variation between participants;
- variation between learning contexts.

3.12 SELF-EVALUATION CRITERIA

The proposed representation will be considered successful if it demonstrates:

Criterion 1

A consistent improvement over the conventional aggregated-feature baseline under participant-level evaluation.

Criterion 2

Evidence that interaction states and/or transitions contribute additional predictive information.

Criterion 3

Improved or maintained generalisation across different learning contexts.

Criterion 4

No substantial performance degradation caused by context conditioning.

Criterion 5

Behavioural states that demonstrate sufficient stability and interpretability to be useful as an analytical representation.

The research will report the magnitude and consistency of improvements rather than claiming success from one accuracy value.

3.13 DEVELOPMENT APPROACH

The component will use an iterative development approach.

The ten-month development cycle will include:

1. requirements identification;

2. literature review;

3. research-gap refinement;

4. behavioural schema design;

5. logger development;

6. pilot testing;

7. main data collection;

8. data preparation;

9. exploratory analysis;

10. temporal representation;

11. state discovery;

12. transition modelling;

13. context modelling;

14. baseline development;

15. proposed model development;

16. evaluation;

17. refinement;

18. integration;

19. final analysis;

20. documentation.

An iterative approach is appropriate because the final behavioural representation cannot be completely determined before the characteristics of the collected data are understood.

3.14 ETHICAL AND PRIVACY CONSIDERATIONS

The TAF specifies informed consent, anonymisation/pseudonymisation, secure local storage, restricted access and the right of withdrawal for collected data. It also states that raw interaction data should not be sent to external servers.

For Component 1:

- raw key characters will not be recorded;
- passwords will not be recorded;
- message contents will not be recorded;
- clipboard contents will not be recorded;
- participant identities will be pseudonymised;
- access to raw logs will be restricted;
- data will be securely stored;
- participants will be informed about the collected information;
- participants will have the right to withdraw according to the approved protocol.

The behavioural logger will therefore capture metadata rather than content.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Component Architecture

The architecture should show the distinction between measurement and the research contribution.

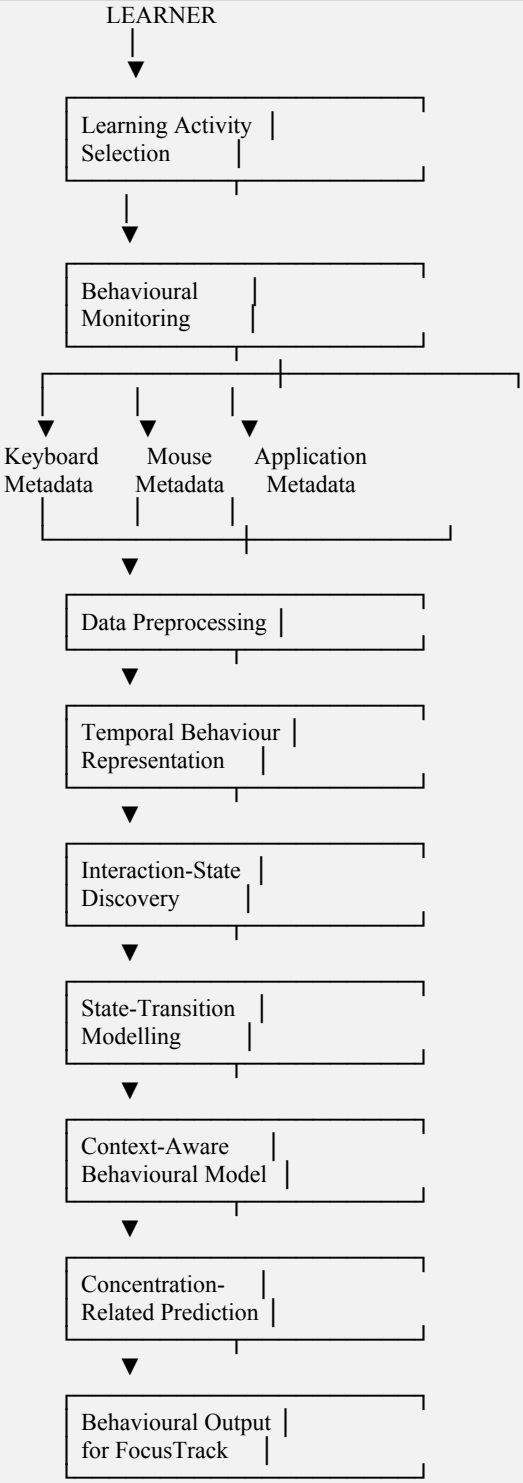


Figure 1: High-Level Component Architecture

Important: According to the proposal template, the final technical architecture diagram should be recreated manually in the required black-and-white format and should not be generated using AI tools.

4.2 INTEGRATION WITH FOCUSTRACK

Component 1 will provide a behavioural information stream to the overall FocusTrack system.



Figure 2: FocusTrack Integration Architecture

Component 1 will provide:

- temporal behavioural features;
- interaction states;
- state durations;
- state transitions;
- interruption/recovery measures;
- context-conditioned behavioural features;
- concentration-related predictions.

The component will not implement environmental sensing or computer vision.

4.3 INDIVIDUAL RESPONSIBILITY BOUNDARY

Activity / Deliverable	My Responsibility	Shared	Other Member
Behavioural literature review	✓		
Behavioural research gap	✓		
Behavioural data specification	✓		
Behaviour logger	✓		
Behavioural preprocessing	✓		
Feature engineering	✓		

Activity / Deliverable	My Responsibility	Shared	Other Member
Interaction-state discovery	✓		
State-transition modelling	✓		
Context-aware behavioural modelling	✓		
Behavioural ML model	✓		
Behavioural evaluation	✓		
Behavioural documentation	✓		
Overall integration		✓	
Final multimodal model		✓	
Environmental sensing			Other member
Computer vision			Other member
Temporal/XAI/decision layer			Other member

This boundary is consistent with the TAF's four-component architecture, where Component 1 is responsible for behavioural analytics while environmental sensing, computer vision and temporal/XAI functions belong to other components.

5. USER REQUIREMENTS

5.1 Requirements-Elicitation Evidence

The current proposal contains preliminary informal elicitation involving:

- semi-structured interviews with 8 undergraduate students;
- an informal survey involving 22 respondents.

The current evidence indicates:

- 18/22 respondents abandoned manual tracking because of tedious entry;
- participants distinguished coding, reading and revision;
- 20/22 ranked minimal interruption highly;
- participants considered simple single-metric summaries insufficient;
- 6/8 expressed distrust of unexplained focus scores;
- privacy/keylogging concerns were raised by 17/22 survey respondents and all 8 interviewees.

These findings should be presented as preliminary proposal-stage evidence, not as population-level research findings.

Formal requirements elicitation should follow the approved research process.

5.2 PROPOSED USER REQUIREMENTS

UR1 — Automatic Monitoring

Students should be able to start a learning session without manually recording every behavioural activity.

Evidence: 18/22 preliminary survey respondents reported abandoning manual tracking because of tedious entry.

UR2 — Learning Activity Selection

Students should be able to specify the learning activity being performed.

Evidence: preliminary interviews identified differences between coding, reading and examination revision.

UR3 — Minimal Interruption

Behavioural monitoring should not significantly interrupt normal learning.

Evidence: 20/22 respondents ranked “does not get in the way” among their top priorities.

UR4 — Pattern-Based Information

The system should provide concentration-related information based on behavioural patterns rather than isolated interaction events.

Evidence: preliminary interview participants considered single activity metrics insufficient without contextual information.

UR5 — Understandable Results

The system should provide understandable behavioural information supporting concentration-related results.

Evidence: 6/8 interviewees reported that they would distrust an unexplained concentration score.

UR6 — Privacy

Students should be able to complete sessions without exposing private information such as passwords or message contents.

Evidence: privacy/keylogging concerns were raised during the preliminary elicitation.

5.3 FUNCTIONAL REQUIREMENTS

FR1: The system shall allow the learner to select a learning activity before starting a session.

FR2: The system shall create a unique session identifier.

FR3: The system shall record timestamped keyboard interaction metadata.

FR4: The system shall record timestamped mouse interaction metadata.

FR5: The system shall identify active application changes.

FR6: The system shall preprocess collected interaction events.

FR7: The system shall generate temporal behavioural features.

FR8: The system shall construct behavioural sequences.

FR9: The system shall identify recurring interaction states.

FR10: The system shall identify transitions between interaction states.

FR11: The system shall associate behavioural representations with learning context.

FR12: The system shall generate concentration-related behavioural predictions.

FR13: The system shall generate session-level behavioural summaries.

FR14: The system shall store processed outputs using the agreed FocusTrack schema.

FR15: The system shall provide behavioural outputs for multimodal integration.

5.4 NON-FUNCTIONAL REQUIREMENTS

Requirement	Proposed Requirement
Privacy	No raw key characters, passwords, message contents or clipboard contents shall be stored.
Security	Behavioural data shall use pseudonymous participant identifiers and restricted access.
Reliability	An individual malformed event shall not terminate the complete session.
Usability	After session setup, the learner should not need to manually record behavioural

Requirement	Proposed Requirement
	events.
Performance	Monitoring should operate without materially interrupting normal learning activity.
Scalability	Additional sessions and participants should be supported without redesigning the complete pipeline.
Reproducibility	Feature definitions, preprocessing, model configurations and evaluation splits shall be documented and version-controlled.
Accessibility	User-facing behavioural summaries should use clear language and readable visual presentation.

6. COMMERCIALIZATION PLAN

6.1 Target Market

Primary Market

- university students;
- higher-education institutions;
- academic learning-support services.

Secondary Market

- EdTech companies;
- learning-analytics providers;
- educational research institutions;
- student productivity platforms.

6.2 CUSTOMER PERSONAS

Student Persona

A university student who wants to understand:

- how their study behaviour changes;
- which learning activities are associated with stronger concentration;
- when interruptions occur;
- how different study sessions compare.

Institutional Persona

A university or educational organisation interested in supporting students using learning analytics while respecting privacy.

6.3 VALUE PROPOSITION

The proposed component can provide more meaningful behavioural information than simple computer-usage totals.

Instead of only reporting:

"You were active for two hours."

the system could provide a contextual behavioural summary describing:

- recurring interaction patterns;
- periods of sustained interaction;
- transitions between behavioural states;
- interruptions;
- recovery patterns;
- differences between learning contexts.

The behavioural component therefore contributes:

- context-aware behavioural analysis;
- temporal behavioural summaries;
- low-cost monitoring;
- minimal manual interaction;
- privacy-aware data collection;
- integration with multimodal FocusTrack analysis.

6.4 REVENUE MODEL

Potential models include:

Freemium

Basic student functionality could be offered at no cost, with advanced analytics provided through a premium service.

Institutional Licensing

Universities could license the platform for student-support or learning-analytics use.

Research Licensing

The analytical framework could potentially be licensed for educational research.

These models are potential commercialization options, not guaranteed revenue outcomes.

6.5 PRICING STRATEGY

Package	**Target**
Free	Individual students
Student Premium	Advanced individual users

Package	Target
Institutional	Universities
Research	Research organisations

Actual pricing should only be determined after:

- market validation;
- cost analysis;
- user testing;
- privacy assessment;
- competitor analysis.

6.6 COMPETITIVE ADVANTAGE

Potential advantages include:

1. context-aware behavioural interpretation;
2. temporal interaction-state modelling;
3. state-transition analysis;
4. low-cost data collection;
5. non-invasive behavioural monitoring;
6. privacy-aware logging;
7. integration with environmental and visual signals.

The strongest potential differentiator is not simply the use of keyboard and mouse data, but the investigation of how those signals can be structured and interpreted according to learning context.

6.7 INTELLECTUAL PROPERTY CONSIDERATIONS

Potential intellectual property may include:

- software implementation;
- behavioural data-processing pipeline;
- interaction-state representation;
- transition-modelling methodology;
- context-aware modelling methodology;
- FocusTrack integration architecture.

Formal patentability or other protection should only be assessed after implementation and novelty evaluation.

7. BUDGET AND JUSTIFICATION

Resource	Estimated Cost	Justification
Existing laptop/PC	Existing resource	Behavioural logging, development and ML
Python	Free	Data collection and modelling
Pandas / NumPy	Free	Data processing
Scikit-learn	Free	Machine Learning
XGBoost	Free/open-source	Candidate ML model
Git	Free	Version control
Local database	Free/open-source	Behavioural storage
Cloud services	Optional	Prototype deployment if required
Additional hardware	Minimal / none	Not required for core behavioural component
Internet	Existing resource	Development and deployment

The component is primarily software-based. Therefore, additional hardware requirements are expected to be low.

The major resource requirement is research and development time across the ten-month period.

8. WORK BREAKDOWN STRUCTURE

Because the project will run for 10 months, the previous 20-week WBS should be replaced.

Ten-Month WBS

WBS	Task	Deliverable	Time
1.1	Literature review	Literature matrix	Month 1
1.2	Research gap refinement	Final research gap	Month 1
1.3	Research questions	RQs	Month 1
2.1	Requirements elicitation	Requirements evidence	Month 2
2.2	Behaviour schema design	Data schema	Month 2
2.3	Privacy design	Privacy specification	Month 2
3.1	Logger architecture	Logger design	Month 2
3.2	Logger development	Behaviour logger	Month 3
3.3	Logger testing	Tested logger	Month 3
4.1	Pilot study preparation	Pilot protocol	Month 4
4.2	Pilot data collection	Pilot dataset	Month 4
4.3	Pilot analysis	Pilot findings	Month 4
5.1	Main data collection	Main dataset	Month 5
5.2	Data quality monitoring	Quality report	Month 5
6.1	Data cleaning	Clean dataset	Month 6
6.2	Feature engineering	Feature dataset	Month 6
6.3	Sequence construction	Behaviour sequences	Month 6
7.1	Exploratory analysis	EDA results	Month 7
7.2	State discovery	Interaction-state model	Month 7
7.3	State validation	State analysis	Month 7
8.1	Transition modelling	Transition model	Month 8
8.2	Context modelling	Context-aware representation	Month 8
8.3	Behavioural interpretation	Context analysis	Month 8
9.1	Baseline development	Baseline model	Month 9
9.2	Proposed model	Proposed ML model	Month 9
9.3	Model evaluation	Evaluation results	Month 9
9.4	Ablation analysis	Ablation results	Month 9

WBS	Task	Deliverable	Time
9.5	Cross-context validation	Generalisation results	Month 9
10.1	Statistical analysis	Statistical results	Month 10
10.2	FocusTrack integration	Integrated component	Month 10
10.3	Final testing	Test report	Month 10
10.4	Final research analysis	Research findings	Month 10
10.5	Final documentation	Final report	Month 10

9. GANTT CHART

The official template expects the Gantt chart to cover literature work, research design, data preparation, experiments, development, prototype work, testing, integration, evaluation, analysis and report preparation.

Use the following 10-month plan for the final Gantt figure.

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Literature review	X	X								
Research-gap refinement	X	X								
Research design	X	X								
Requirements elicitation		X								
Behaviour schema		X								
Logger development		X	X							
Prototype testing			X	X						
Pilot data collection				X						
Main data collection					X					
Data preparation				X	X	X				
Exploratory analysis				X	X	X	X			
Temporal representation						X	X			
State discovery							X			
Transition modelling							X	X		
Context modelling								X		
Baseline development									X	

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Proposed model development									X	
Experiments							X	X	X	
Ablation analysis									X	
Cross-context validation									X	
Statistical analysis										X
Integration										X
Final testing										X
Final analysis										X
Report preparation								X	X	X

Ten-Month Logic

The work is intentionally iterative:

```
M1–M2: Research foundation
      ↓
M3–M4: Prototype and pilot
      ↓
M5: Main data collection
      ↓
M6: Data preparation
      ↓
M7–M8: Behavioural state and transition research
      ↓
M9: ML experiments and validation
      ↓
M10: Statistical analysis, integration and final evaluation
```

This is much more appropriate for a 10-month research project than compressing the complete component into 20 weeks.

10. REFERENCES

The current proposal contains the following references. The official template explicitly states that students remain responsible for verifying references, including references identified with AI assistance.

[1] K. Altuwairqi, S. Kammoun Jarraya, A. Allinjawi, and M. Hammami, “Student behavior analysis to measure engagement levels in online learning environments,” Signal, Image and Video Processing, vol. 15, pp. 1387–1395, 2021.

[2] D. Biedermann, G.-P. Ciordas-Hertel, M. Winter, J. Mordel, and H. Drachsler, “Contextualized Logging of On-Task and Off-Task Behaviours During Learning,” Journal of Learning Analytics, vol. 10, no. 2, pp. 115–125, 2023.

[3] G.-E. Cha and B.-C. Min, “Correlation between Unconscious Mouse Actions and Human Cognitive Workload,” in CHI Conference on Human Factors in Computing Systems Extended Abstracts, 2022.

[4] P. Freihaut and A. S. Göritz, “Using the computer mouse for stress measurement – An empirical investigation and critical review,” International Journal of Human-Computer Studies, vol. 145, Art. no. 102520, 2021.

[5] J. Jovanović, M. Saqr, S. Joksimović, D. Gašević, and A. Kurdi, “Students matter the most in learning analytics: The effects of internal and instructional conditions in predicting academic success,” Computers & Education, vol. 172, Art. no. 104251, 2021.

[6] A. N. Meyer et al., “Detecting Developers' Task Switches and Types,” IEEE Transactions on Software Engineering, vol. 48, pp. 225–240, 2022.

[7] H. Yakura, T. Nakano, and M. Goto, “An automated system recommending background music to listen to while working,” User Modeling and User-Adapted Interaction, vol. 32, pp. 355–388, 2022.

[8] E. Cavallin and S. Spagnol, “When Designers Sweat: Behavioral Traces of GenAI Co-Creation,” in Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems, 2026.

[9] E. S. Debie et al., “Multimodal Fusion for Objective Assessment of Cognitive Workload: A Review,” IEEE Transactions on Cybernetics, vol. 51, pp. 1542–1555, 2021.

Important: Before final submission, verify each bibliographic entry against the original publication, as required by the proposal instructions.

11. APPENDICES

APPENDIX A — PROPOSED BEHAVIOURAL DATA SCHEMA

Field	Description
participant_id	Anonymous participant identifier
session_id	Learning-session identifier
timestamp	Event timestamp
learning_context	Selected learning activity
event_type	Keyboard / mouse / application event
application_category	Active application category
duration	Duration of activity/window
interaction_state	Discovered behavioural state
state_transition	Transition between states
concentration_label	Session-level reference label

Privacy Rule

The schema shall not contain:

- raw key characters;
- passwords;
- private messages;

- clipboard contents.

APPENDIX B — RESEARCH DECISION LOG

Claim / Problem	Evidence	Decision
Interaction data contains useful behavioural information	[1], [3], [4]	Use keyboard, mouse and application metadata
Behaviour can be ambiguous	[2], [4]	Avoid single-feature interpretation
Learning context affects interpretation	[2], [5]	Include learning activity
Behaviour changes over time	[6]	Use temporal representation
Interaction can contain recurring phases	[8]	Investigate state discovery
State changes contain useful information	[6]	Model transitions
Behavioural modality can complement other modalities	[9]	Integrate with FocusTrack
Proposed representation requires empirical proof	Research-design requirement	Compare against baseline

APPENDIX C — AI USE DISCLOSURE

This appendix should reflect the actual use of AI tools during proposal preparation.

AI Tool	Purpose	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
Asta Allen AI (asta.allen.ai)	Search Research Papers	Assisted with finding potentially relevant academic papers and literature for the literature review and research-gap analysis.	The suggested papers and information were checked against the original academic publications before being used in the report.	Independently selected relevant sources, read and evaluated the original papers, interpreted the findings, identified the research gap, and wrote the final literature review and research content.
OpenCode	Assist with implementation of the data-collection software for the behavioural data component.	Assisted with generating, modifying, debugging, and explaining code used for collecting and processing	The code was reviewed, tested, debugged, and validated by running the data-collection system and checking	The student defined the data-collection requirements, selected the variables to be collected,

		behavioural data during research sessions.	whether the collected data matched the defined requirements.	reviewed and modified the code, conducted testing, and remained responsible for the final implement
ChatGPT	Assist with report preparation, writing refinement, structuring, and clarification of research ideas.	Assisted with improving wording, organizing sections, explaining concepts, and refining the presentation of researcher-developed content.	The generated suggestions were reviewed, corrected, and compared with the research materials, academic sources, and project requirements.	The student made the research decisions, selected the content, verified the information, and was responsible for the final report and its academic content.

The official proposal instructions state that AI use does not transfer responsibility from the student. The student remains responsible for accuracy, originality, reference validity, research decisions, methodology and interpretation.

Do not copy this table blindly. Replace it with your actual AI-use history.

APPENDIX D — REQUIREMENTS ELICITATION EVIDENCE

The current preliminary elicitation consists of semi-structured interviews with 8 undergraduate students and an informal survey involving 22 respondents.

Question	Finding	Requirement
Have you used a time-tracking/study-log tool before?	18/22 reported abandoning manual tools because of tedious entry	UR1
Does learning activity affect how behaviour should be tracked?	Coding, reading and revision were consistently distinguished	UR2
What is important in a monitoring tool?	20/22 ranked minimal interruption highly	UR3
What do you think about simple activity summaries?	Single metrics were considered insufficient without context	UR4
Would you trust an automatic focus score?	6/8 wanted supporting explanation	UR5
What concerns would prevent use?	Privacy/keylogging concerns raised by 17/22 and 8/8	UR6

Important: This is preliminary proposal-stage evidence. It should not be presented as statistically representative of university students. The formal requirements-elicitation process should follow the approved research procedure.

FINAL RESEARCH POSITION OF COMPONENT 1

Your component is now much clearer academically.

The TAF originally describes the component as:

behavioural classification using keyboard, mouse and application interaction.

The revised proposal keeps those approved inputs and ML technologies, but moves the research contribution one level deeper:

